SSP

Computer & Communication Department

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Modern Physics
Sheet 5
Solutions

Eng. Mohammad Alkammash

1. Show that the phenomenon of time dilation is contained in the Lorentz coordinate transformation. A light located at (x_0, y_0, z_0) is turned abruptly on at t_1 and off at t_2 in frame S. (a) For what time interval is the light measured to be on in frame S'. (b) What is the distance between where the light is turned on and off as measured by S'?

Solution

- Let the two events happen in the same place in frame S. (Frame S measures the proper time)

	Event 1	Event 2
Frame S	x_0, t_1	x_0, t_2
Frame S'	x_1', t_1'	x_2', t_2'

$$t_1' = \gamma(t_1 - \frac{vx_0}{c^2})$$

$$t_2' = \gamma(t_2 - \frac{vx_0}{c^2})$$

$$t_2' - t_1' = \gamma\left(t_2 - \frac{vx_0}{c^2}\right) - \gamma\left(t_1 - \frac{vx_0}{c^2}\right) = \gamma(t_2 - t_1)$$

$$\Delta t' = \gamma \Delta t$$

2. Use the Lorentz transformation to derive the expression for length contraction. Note that the length of a moving object is determined by measuring the positions of both ends simultaneously.

Solution

Let the object be at rest in frame S. (S measures the proper length)

	Event 1	Event 2
Frame S	x_1, t_1	x_2, t_2
Frame S'	x_1', t_1'	x_2', t_1'

$$x_1' = \gamma(x_1 - vt_1)$$

$$x_2' = \gamma(x_2 - vt_2)$$

$$x_2' - x_1' = \gamma(x_2 - vt_2) - \gamma(x_1 - vt_1)$$

$$= \gamma(x_2 - x_1) - \gamma v(t_2 - t_1) \qquad (1)$$

$$t_1' = \gamma(t_1 - \frac{vx_1}{c^2})$$

$$t_2' = \gamma(t_2 - \frac{vx_2}{c^2})$$

$$t_1' = t_2'$$

$$t_2 - t_1 = \frac{v}{c^2}(x_2 - x_1) \qquad (2)$$

Substitute (2) in (1)

$$x_2' - x_1' = \gamma(x_2 - x_1) - \gamma\frac{v^2}{c^2}(x_2 - x_1) = \gamma(x_2 - x_1)\left(1 - \frac{v^2}{c^2}\right) = \frac{(x_2 - x_1)}{\gamma}$$

$$\Delta x' = \frac{\Delta x}{\gamma}$$

3. An observer in frame *S* sees lightning simultaneously strike two points 100 m apart. The first strike occurs at $x_1 = y_1 = z_1 = t_1 = 0$ and the second at $x_2 = 100\ m$, $y_2 = z_2 = t_2 = 0$. (a) What are the coordinates of these two events in a frame *S'* moving in the standard configuration at 0.70*c* relative to *S*? (b) How far apart are the events in *S'*? (c) Are the events simultaneous in *S'*? If not, what is the difference in time between the events, and which event occurs first?

Solution

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{1}{\sqrt{1-0.7^2}} = 1.40$$

$$x_1' = \gamma(x_1 - vt_1) = 0$$

$$y_1' = y_1 = 0$$

$$z_1' = z_1 = 0$$

$$t_1' = \gamma\left(t_1 - \frac{vx_1}{c^2}\right) = 0$$

$$x_2' = \gamma(x_2 - vt_2) = 1.40(100 - 0) = 140\ m$$

$$y_2' = y_2 = 0$$

$$z_2' = z_2 = 0$$

$$t_2' = \gamma\left(t_2 - \frac{vx_2}{c^2}\right) = 1.40\left(0 - \frac{0.7 \times 100}{3 \times 10^8}\right) = -0.33\ \mu s$$

$$\Delta x' = 140\ m$$

Events are not simultaneous in S′.

Event 2 occurs 0.33 μs earlier than event 1.

4. The spacetime coordinates of two events as measured by S are $x_1 = 6\times10^4$ m, $t_1 = 2\times10^{-4}$ s, $x_2 = 12\times10^4$ m, $t_2 = 1\times10^{-4}$ s. What must be the velocity of S' relative to S if the two events are simultaneous in S'?

Solution

$$t_1' = \gamma\left(t_1 - \frac{vx_1}{c^2}\right)$$

$$t_2' = \gamma\left(t_2 - \frac{vx_2}{c^2}\right)$$

The events are simultaneous in S' so the $t_1' = t_2'$

$$t_1 - \frac{vx_1}{c^2} = t_2 - \frac{vx_2}{c^2}$$

$$t_1 - t_2 = \frac{v}{c^2}(x_1 - x_2)$$

$$\frac{v}{c} = \frac{t_1 - t_2}{x_1 - x_2} \times c = \frac{2 \times 10^{-4} - 1 \times 10^{-4}}{6 \times 10^4 - 12 \times 10^4} \times 3 \times 10^8 = -0.5$$

$$v = -0.5c$$

5. An observer on Earth observes two spacecraft moving in the same direction toward the Earth. Spacecraft A appears to have a speed of 0.50c, and spacecraft B appears to have a speed of 0.80c. What is the speed of spacecraft A measured by an observer in spacecraft B?

Solution

Particle: Spacecraft A

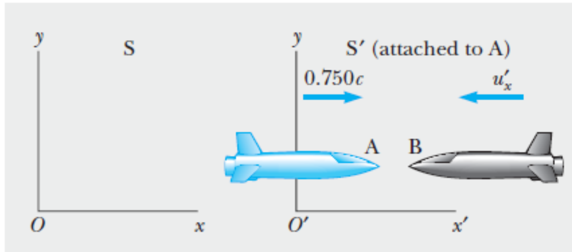
Frame S: Earth

Frame S': Spacecraft B

$u_x = 0.50c$
$v = 0.80\ c$

$$u'_x = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} = \frac{0.50c - 0.80c}{1 - 0.50 \times 0.80} = -0.50c$$

6. Two spaceships A and B are moving in opposite directions as shown in the figure. An observer on Earth measures the speed of A to be $0.750c$ and the speed of B to be $0.850c$. Find the velocity of B with respect to A.



Solution

Particle: Spacecraft B

Frame S: Earth

Frame S/: Spacecraft A

$$u_x = -0.850c$$
$$v = 0.750\ c$$

$$u'_x = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} = \frac{-0.850c - 0.750c}{1 + 0.850 \times 0.750} = -0.977c$$

7. Two spaceships approach each other, each moving with the same speed as measured by an observer on the Earth. If their relative speed is 0.70c, what is the speed of each spaceship?

Solution

Particle: Spacecraft A

Frame S: Earth

Frame S$^{\prime}$: Spacecraft B

$$v = -u_x$$
$$u_x' = 0.7c$$

$$u_x' = \frac{u_x - v}{1 - \frac{u_x v}{c^2}}$$

$$0.7c = \frac{2u_x}{1 + \frac{u_x^2}{c^2}}$$

$$u_x = 0.408\ c$$

8. Imagine a motorcycle rider moving with a speed of 0.800*c* past a stationary observer. If the rider tosses a ball in the forward direction with a speed of 0.700*c* with respect to himself, what is the speed of the ball as seen by the stationary observer?

Solution

Particle: ball

Frame S: motorcycle

Frame S′: stationary observer

$$u_x = 0.700\ c$$

$$v_{S'/s} = -v_{S/S'} = -0.800\ c$$

$$u'_x = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} = \frac{0.700c + 0.800c}{1 + \frac{0.700c \times 0.800c}{c^2}} = 0.962\ c$$

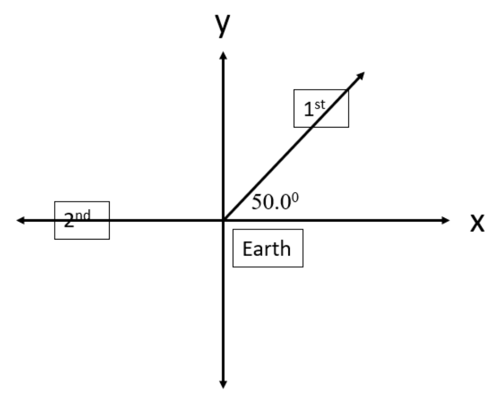
9. Suppose that the motorcyclist moving with a speed $0.800c$ turns on a beam of light that moves away from him with a speed of c in the same direction as the moving motorcycle. What would the stationary observer measure for the speed of the beam of light?

Solution

The speed of light is constant independent of the speed of the observer, so the stationary observer measures the speed of light as c.

10. A spacecraft is launched from the surface of the Earth with a velocity of 0.600c at an angle of 50.0^0 above the horizontal, positive x-axis. Another spacecraft is moving past with a velocity of 0.700c in the negative x direction. Determine the magnitude and direction of the velocity of the first spacecraft as measured by the pilot of the second spacecraft.

Solution



Particle: first spacecraft

Frame S: earth

Frame S': second spacecraft

$$u_x = 0.600c\cos(50)$$

$$u_y = 0.600c\sin(50)$$

$$v = -0.700c$$

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = 1.400$$

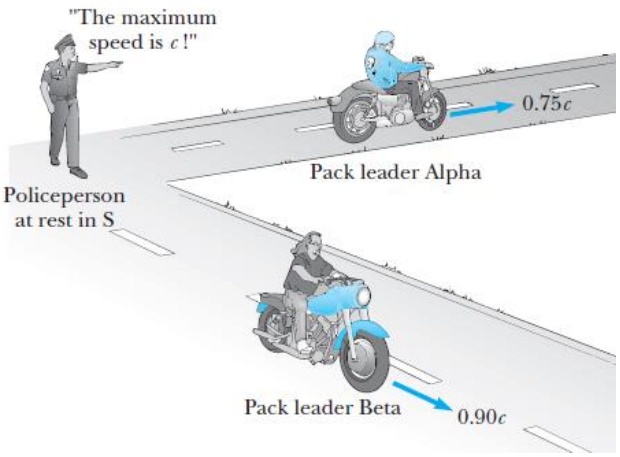
$$u'_x = \frac{u_x - v}{1-\frac{u_x v}{c^2}} = \frac{0.600c\cos(50) + 0.700\,c}{1 + 0.600\cos(50) \times 0.700} = 0.855c$$

$$u'_y = \frac{u_y}{\gamma(1-\frac{u_x v}{c^2})} = \frac{0.600c\sin(50)}{1.400(1 + 0.600\cos(50) \times 0.700)} = 0.259c$$

$$u' = \sqrt{{u'_x}^2 + {u'_y}^2} = \sqrt{(0.855c)^2 + (0.259c)^2} = 0.893c$$

$$\theta = \tan^{-1}\left(\frac{u'_y}{u'_x}\right) = \tan^{-1}\left(\frac{0.259c}{0.855c}\right) = 16.9°$$

11. Imagine two motorcycle gang leaders racing at relativistic speeds along perpendicular paths from the local pool hall ash shown in the figure. How fast does pack leader Beta recede over Alpha's right shoulder as seen by Alpha?



Solution

Particle: Beta

Frame S: earth

Frame $S^{/}$: Alpha

$u_x = 0$

$u_y = -0.90c$

$v = 0.75c$

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = 1.51$$

$$u'_x = \frac{u_x - v}{1-\frac{u_x v}{c^2}} = \frac{0-0.75\,c}{1-0} = -0.75c$$

$$u'_y = \frac{u_y}{\gamma(1-\frac{u_x v}{c^2})} = \frac{-0.90c}{1.51(1-0)} = -0.60c$$

$$u' = \sqrt{{u'_x}^2 + {u'_y}^2} = \sqrt{(-0.75c)^2 + (-0.60c)^2} = 0.96c$$

$$\theta = \tan^{-1}\left(\frac{u'_y}{u'_x}\right) = \tan^{-1}\left(\frac{-0.60c}{-0.75c}\right) = 38.7°$$